

AI HUMAN ACTIVITY RECOGNITION FOR ONBOARD BAS EXPERIMENTS

FINAL ARCHITECTURE, END-TO-END PIPELINE & PROTOTYPE PLAN

Final sensing definition: Fixed RGB camera + Wi-Fi Channel State Information (CSI). **3D Human Mesh Recovery (3D HMR)** is included in the final vision architecture. No separate radar, mmWave, point-cloud, or other sensing subsystem is used.

Single canonical workflow: Fixed Camera → Vision Perception → Hand–Object Interaction → 3D HMR → Vision Features; CSI Capture → CSI Processing → CSI Embedding; Vision + CSI + HMR → Confidence-Aware Feature Fusion → Temporal HAR → Debounce → Experiment FSM → Event Bus → Voice / Structured Log / Video+RTSP / Dashboard.

1. Problem Description

An onboard, offline, experiment-aware Human Activity Recognition (HAR) co-pilot observes an astronaut performing a predefined experiment, recognizes actions and object interactions, reasons over their temporal order, detects protocol violations, guides the astronaut and maintains a lightweight digital record.

1.1 Core Challenges

- Microgravity removes reliable floor/up/down assumptions.
- Single frames are ambiguous; temporal context is required.
- ML recognition and protocol validation must be separate.
- Wi-Fi CSI provides complementary motion/context evidence when vision is uncertain.
- The real-time loop operates locally without continuous Earth communication.

1.2 Representative Experiment

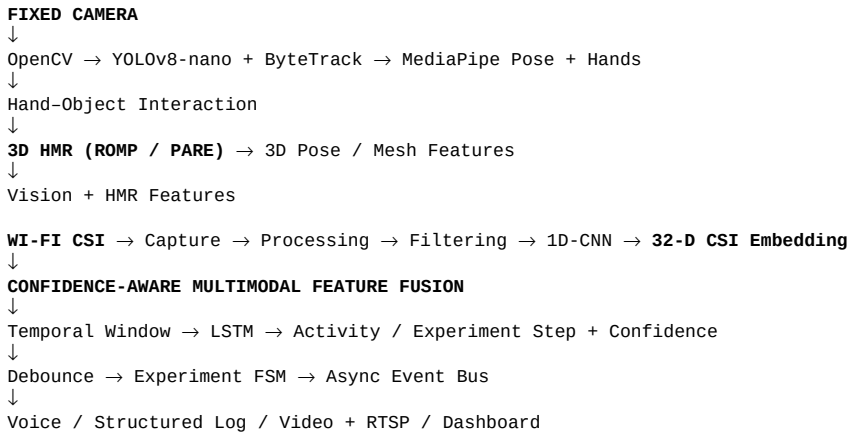
Open container → Pick red object → Place red object → Pick yellow object → Place yellow object → Close container → Complete.

If PICK_YELLOW occurs while PLACE_RED is expected, the FSM flags the violation and the system provides corrective guidance.

2. Final Architecture

The fixed camera is the primary semantic sensing path. OpenCV acquires frames; YOLOv8-nano detects experiment objects and ByteTrack maintains identities; MediaPipe Pose and Hands estimate body/hand landmarks; hand-object geometry determines interaction states; 3D HMR provides an additional 3D human pose/mesh representation.

In parallel, Wi-Fi CSI is captured independently, processed and encoded into a 32-D embedding. The vision representation, HMR representation and CSI embedding are brought together only at the multimodal fusion stage.



3. Architecture Components

Stage	Technology	Primary Output
Camera	Fixed RGB camera + OpenCV	Frames
Object perception	YOLOv8-nano + ByteTrack	Objects, boxes, confidence, track IDs
Human perception	MediaPipe Pose + Hands	Body, wrist and hand landmarks
Interaction	Hand-object distance / IoU / geometric reasoning	NONE / NEAR / GRASPING
3D human representation	3D HMR: ROMP / PARE	3D pose / mesh features
CSI sensing	Wi-Fi CSI + Nexmon CSI	CSI stream
CSI processing	Filtering + normalization + 1D-CNN	32-D CSI embedding

Stage	Technology	Primary Output
Fusion	Confidence-aware multimodal fusion	Fused representation
Temporal HAR	2-layer LSTM, 30-frame window	Step + confidence
Stabilization	Debounce / stable evidence	Confirmed step
Protocol	JSON-defined FSM	OK / violation / done
Outputs	pyttsx3; JSON/TXT; FFmpeg/RTSP; FastAPI/WebSocket	Guidance / log / video / dashboard

4. End-to-End Pipeline

- 1. Capture:** Fixed camera continuously captures frames through a bounded queue.
- 2. Vision perception:** YOLOv8-nano detects experiment objects; ByteTrack maintains object identity; MediaPipe supplies body and hand landmarks.
- 3. Hand-object interaction:** Hand-to-object geometry converts detections and landmarks into NONE/NEAR/GRASPING-style interaction states.
- 4. 3D HMR:** ROMP/PARE processes RGB frames to produce 3D human pose/mesh features. These are included in the vision feature representation.
- 5. CSI capture:** Wi-Fi CSI is acquired independently, for example using Nexmon CSI, at the configured sampling rate.
- 6. CSI processing:** CSI features are extracted, filtered and encoded with a 1D-CNN into a compact 32-D embedding.
- 7. Feature fusion:** Vision + HMR features and CSI embedding are fused with confidence awareness.
- 8. Temporal HAR:** A 30-frame sliding window is passed to the LSTM to infer the candidate experiment step and confidence.
- 9. Debounce:** A step is confirmed only after stable evidence; transient predictions do not advance the protocol.
- 10. FSM validation:** The confirmed action is checked against the expected action in the JSON-defined experiment state machine.
- 11. Event dispatch:** An asynchronous event bus distributes the result without blocking inference.
- 12. Response:** Correct progression triggers next-step guidance; violations trigger corrective voice alerts; logs, dashboard and video are updated.

5. Vision Perception Pipeline

5.1 Object Detection and Tracking

Component	Specification
Detector	YOLOv8-nano
Framework	Ultralytics
Training	Fine-tune on actual experiment tools/objects
Tracking	ByteTrack
Outputs	Class, bounding box, confidence, track ID, presence/position
Optimization	ONNX / INT8 after functional validation

5.2 Pose and Hands

Component	Specification
Body pose	MediaPipe Pose
Hands	MediaPipe Hands
Alternative	MediaPipe Holistic
Outputs	Body landmarks, wrist keypoints, hand landmarks
Hand representation	21 landmarks/hand; 126-D coordinate representation in the base feature vector

5.3 Hand–Object Interaction

The interaction layer connects human pose with object semantics. Proximity, overlap and hand-to-object geometry distinguish generic reaching from object-specific interaction.

Method	Use
IoU	Bounding-box overlap / association
Keypoint-to-BBox distance	Hand-object association
Hand-object distance	Proximity reasoning

Method	Use
State	NONE / NEAR / GRASPING

6. 3D Human Mesh Recovery

Component	Final Design
3D HMR	Included in final architecture
Candidate frameworks	ROMP / PARE
Input	RGB camera frames
Output	3D human body pose, 3D joints, human mesh parameters
Reference representation	Payload-rack-relative coordinates
Purpose	3D human representation for orientation-aware pose reasoning
Position	After hand-object interaction and before multimodal fusion
Temporal integration	3D pose/mesh features are available to the temporal HAR representation
Deployment	GPU-oriented; Jetson-class target
Reference datasets	AGORA / 3DPW

Feature-dimension rule: The supplied architecture specifies a 147-D base vision representation and a 32-D CSI embedding. Because the exact HMR projection dimension is not specified in the source design, the final fused dimension is expressed as $179 + D_{\text{HMR}}$. The LSTM input must use that same final dimension. No unsupported fixed HMR dimension is assumed.

7. CSI Complementary Sensing

CSI means Wi-Fi Channel State Information. It is the only complementary sensing modality in this final design.

Stage	Specification
Acquisition	Nexmon CSI / CSI-capable Wi-Fi setup
Node	Raspberry Pi 5 or equivalent CSI-capable node
Target sampling	~100 Hz
Transport	UDP
Buffer	Rolling CSI buffer
Synchronization	Camera/CSI timestamp alignment
Processing	Feature extraction + normalization + filtering
Reference filter	0.5–10 Hz
Encoder	PyTorch 1D-CNN
Embedding	32-D CSI embedding

CSI role: CSI provides movement/presence and motion-pattern evidence and improves robustness when visual information becomes unreliable. It does not inherently identify which object was manipulated; fine-grained object semantics remain primarily visual.

Terminology boundary: This implementation is not described as a separate radar, mmWave, point-cloud or other non-CSI sensing system.

8. Multimodal Feature Representation

Feature	Reference Dimension
Object presence	4
Object positions	6
Wrist keypoints	6
Hand landmarks	126
Interaction state	5
Base vision representation	147
CSI embedding	32
Base camera + CSI representation	179
3D HMR	D_{HMR} — selected projection/output dimension
Final fused representation	$179 + D_{\text{HMR}}$
Temporal window	30 frames

The earlier 179-D value remains valid only for the base camera/interaction + CSI representation. Once 3D HMR is included in the actual fused representation, the dimension must increase by the selected HMR feature dimension.

9. Confidence-Aware Feature Fusion

Condition	System Behavior
Normal	Vision + HMR + CSI evidence
Visual uncertainty	CSI provides complementary motion/context evidence
CSI unavailable	Camera + HMR degraded mode
Both uncertain	Hold / abstain
Object semantics	Remain primarily vision-derived
Fusion point	After independent camera/HMR and CSI processing

10. Temporal Human Activity Recognition

Parameter	Reference Design
Framework	PyTorch 2.x
Model	2-layer LSTM
Hidden size	256
Sequence length	30 frames
Input size	179 + D_HMR
Temporal context	~1 second at 30 FPS
Classification head	256 → 128
Activation	ReLU
Dropout	0.3
Output	Candidate experiment step + confidence

The temporal model uses sequence context to distinguish similar instantaneous poses by their trajectories over time.

11. Prediction Stabilization / Debounce

Mechanism	Reference
Confidence threshold	>0.75
Temporal debounce	Enabled
Stable confirmation	15-frame reference
Transient prediction	Does not advance protocol
Low confidence	Hold state
Insufficient evidence	Abstain
Transition gating	Required

12. Experiment Protocol Validation

Current State	Expected Next State
START	OPEN_CONTAINER
OPEN_CONTAINER	PICK_RED
PICK_RED	PLACE_RED
PLACE_RED	PICK_YELLOW
PICK_YELLOW	PLACE_YELLOW
PLACE_YELLOW	CLOSE_CONTAINER
CLOSE_CONTAINER	DONE

The ML model proposes the observed action. The deterministic JSON-defined FSM decides whether that action is legal in the current protocol state.

Event	Response
STEP_OK	Advance and announce next step
STEP_SKIPPED	Violation + corrective voice guidance
OUT_OF_ORDER	Violation + corrective voice guidance
INVALID_TRANSITION	Reject transition
LOW_CONFIDENCE	Hold state
EXPERIMENT_DONE	Close session + final log

13. Event Management

Component	Technology / Role
Event architecture	Asynchronous event bus
Runtime	asyncio
Dispatch	Non-blocking
Consumers	Voice, logger, video storage, RTSP, dashboard, WebSocket, status monitor

14. Outputs

Output	Technology
Voice guidance	pyttsx3, offline TTS
Structured logging	JSON / CSV / TXT
Local video	OpenCV VideoWriter / MP4
Streaming	FFmpeg / RTSP
Alternative streaming	GStreamer
Optional RTSP server	MediaMTX
Dashboard backend	FastAPI
Live transport	WebSocket
Frontend	HTML / JavaScript
Prototype GUI	Streamlit / PyQt5

15. Dashboard Information

- Current experiment
- FSM state
- Current step
- Confidence
- Progress
- Alerts
- Protocol status
- Camera status
- CSI status
- Degraded sensing mode
- Optional live video

16. Data and Training Strategy

Data	Purpose
Self-collected fixed-camera images	Object detection
Complete experiment sequences	Temporal HAR
Correct runs	Normal execution
Step skips	Violation recognition
Wrong-object actions	Semantic error handling
Pauses	Temporal robustness
Occlusions	Visual degradation testing
Lighting changes	Vision robustness
Synchronized CSI sequences	CSI and multimodal training
Held-out sessions	Generalization evaluation

17. Datasets

Dataset / Source	Use
COCO	Object detection reference/pretraining

Dataset / Source	Use
Open Images	Object detection reference
Roboflow	Dataset management / annotation
COCO-WholeBody	Whole-body / hand reference
AGORA	3D HMR
3DPW	3D human pose/mesh
EPIC-KITCHENS	Human-object interaction reference
HOI4D	Human-object interaction reference
Custom synchronized camera + CSI dataset	Primary multimodal training/evaluation

18. Deployment Architecture

Component	Design
Primary edge target	NVIDIA Jetson Orin/Xavier-class
Prototype fallback	Laptop / Intel NUC
CSI node	Raspberry Pi-class or equivalent
Runtime	Decoupled camera, CSI, inference, audio, logging and dashboard/streaming processes
Optimization	ONNX, ONNX Runtime, INT8
Advanced optimization	TensorRT
3D HMR	GPU-oriented

Final latency must be measured on the actual demonstration hardware; design estimates are not treated as measured guarantees.

19. Runtime Concurrency

Thread / Process	Responsibility
Camera thread	Camera acquisition + bounded frame queue
CSI thread	CSI packet reception + embedding buffer
Main inference	Vision + HMR + interaction + fusion + LSTM + FSM
Audio thread	Voice playback
I/O thread	Session logging
Stream thread	Video / RTSP
Async server	FastAPI / WebSocket

20. Failure Handling

Failure	Response
Camera uncertainty	CSI supporting evidence
CSI unavailable	Camera + HMR degraded mode
Wrong object	Reject through interaction + FSM
Skipped action	FSM violation + voice correction
Out-of-order action	FSM violation + voice correction
Both uncertain	Hold / abstain
Low confidence	No state transition
Experiment complete	Close session + final record

21. Prototype Build Plan

- Freeze a 5–7 step experiment protocol.
- Build the fixed-camera mock testbed and collect complete sequences.
- Train/validate YOLOv8-nano on actual experiment tools.
- Add MediaPipe Pose + Hands and hand-object interaction.
- Integrate 3D HMR (ROMP/PARE) and define the selected HMR feature projection.
- Implement CSI capture and 32-D CSI embedding.
- Build the fused representation and temporal LSTM using the resulting final feature dimension.

- Implement debounce and FSM validation.
- Add voice, structured logs, dashboard and video.
- Demonstrate CSI-assisted robustness under visual uncertainty.
- Optimize the complete pipeline for edge hardware.

22. Judge Demonstration

- Correctly execute several experiment steps and show next-step guidance.
- Deliberately skip PLACE_RED and attempt PICK_YELLOW; show FSM violation and corrective alert.
- Introduce visual occlusion/confidence degradation; show CSI continuing to provide motion/context evidence.
- Show the 3D HMR pose/mesh representation as part of the perception layer.
- Restore visual confidence and complete the experiment.
- Show the structured log and dashboard.

23. Core Innovation

- Experiment-aware temporal HAR.
- Probabilistic perception + deterministic protocol validation.
- Confidence-aware fusion of camera/HMR and Wi-Fi CSI evidence.
- 3D human pose/mesh representation for rack-relative orientation-aware reasoning.
- Offline edge-first operation.
- Proactive next-step experiment guidance.

24. Single Source of Truth — Final Workflow

VISION BRANCH

Fixed Camera → OpenCV → YOLOv8-nano + ByteTrack → MediaPipe Pose + Hands → Hand-Object Interaction → 3D HMR (ROMP/PARE) → Vision/HMR Features

CSI BRANCH

Wi-Fi CSI Capture → CSI Processing → 0.5-10 Hz Filtering → 1D-CNN → 32-D CSI Embedding

FUSION + DECISION

Vision/HMR + CSI → Confidence-Aware Feature Fusion → 30-Frame LSTM → Debounce → Experiment FSM → Async Event Bus

OUTPUTS

Voice / Structured Log / Video + RTSP / Dashboard

25. Final Architecture Statement

The final proposed system is an offline, edge-deployed, experiment-aware HAR co-pilot. A fixed RGB camera feeds object detection/tracking, body and hand estimation, hand-object interaction and 3D human mesh recovery. In parallel, Wi-Fi CSI is captured and processed into a compact 32-D motion embedding. The vision/HMR and CSI representations are fused using confidence-aware multimodal reasoning, accumulated over a 30-frame temporal window and classified by an LSTM into an experiment step and confidence. Debounce confirms stable actions, after which a JSON-defined finite-state machine validates the action against the expected protocol state. Valid transitions advance the experiment and trigger next-step guidance; violations generate corrective offline voice alerts. An asynchronous event bus distributes events to structured logging, local video/RTSP and a FastAPI/WebSocket dashboard.

26. Final Design Boundaries

Included: Fixed RGB camera, OpenCV, YOLOv8-nano, ByteTrack, MediaPipe Pose/Hands, hand-object interaction, 3D HMR (ROMP/PARE), Wi-Fi CSI, CSI processing, 1D-CNN CSI embedding, confidence-aware fusion, LSTM, debounce, FSM, event bus, voice, logging, video, RTSP and dashboard.

Not included: Any separate radar, mmWave, point-cloud, or other non-CSI sensing subsystem.

Consistency rule: The architecture uses one workflow everywhere. Camera/HMR and CSI remain separate until feature fusion. The old 179-D value is not claimed as the final HMR-inclusive dimension; the correct final dimension is 179 + D_HMR unless a concrete HMR projection is selected.